

While loops

“For” loops are useful when you know how many times a task needs to be repeated. But sometimes you’ll need a loop to keep repeating until something changes. A “while” loop keeps on going around as many times as it needs to.

SEE ALSO

◀ **118–119** Making decisions

◀ **122–123** Loops in Python

Escaping **126–127** › loops

While loops

A while loop keeps repeating as long as a certain condition is true. This condition is called the “loop condition” and is either true or false.

► How it works

A while loop checks if the condition is true. If it is, it goes around the loop again. If it’s not, it skips the loop.



1 Create a while loop

Set the starting value of the “answer” variable in the loop condition. The loop condition has to be true to start with or the program will never run the loop.

The code inside the loop must be indented four spaces

```

answer = 'y'
while answer == 'y':
    print('Stay very still')
    answer = input('Is the monster friendly? (y/n)')
print('Run away!')
  
```

The “answer” variable is set to “y”

The while loop only runs if the condition is true

If the condition is false, unindented code after the loop runs and a different message appears

2 What the program looks like

The value entered is stored in the variable “answer”. The loop condition is “answer == ‘y’”. If you type “y”, the loop keeps going. If you type “n”, the loop stops.



```
>>>
```

```
Stay very still
```

```
Is the monster friendly? (y/n)y
```

```
Stay very still
```

```
Is the monster friendly? (y/n)y
```

```
Stay very still
```

```
Is the monster friendly? (y/n)n
```

```
Run away!
```

Answer is “y”, so the loop keeps running

Answer is “n”, so the loop ends and a new message appears

REMEMBER

“repeat until” block

Python’s “while” loop is similar to the “repeat until” block in Scratch. Both keep on repeating until something different happens in the program.

